

2.4 Correlation matrices

Covariances carry units: the covariance of income in dollars with age in years is in dollar-years. To compare the strength of linear association across pairs of variables, each covariance is divided by the two standard deviations.

Definition 2.4.1 — Correlation matrix

Let $\mathbf{Y}$ have covariance matrix $\mathbf{\Sigma} = (\sigma_{ij})$ with every $\sigma_{ii} > 0$, and let $\mathbf{D}_\sigma = \text{diag}(\sigma_{11}^{1/2}, \dots, \sigma_{pp}^{1/2})$ be the diagonal matrix of standard deviations. The **correlation matrix** of $\mathbf{Y}$ is

$$\mathbf{R} = \mathbf{D}_\sigma^{-1} \mathbf{\Sigma} \mathbf{D}_\sigma^{-1}, \quad \text{with entries} \quad \rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} = \text{corr}(Y_i, Y_j).$$

Equivalently $\mathbf{\Sigma} = \mathbf{D}_\sigma \mathbf{R} \mathbf{D}_\sigma$. For random vectors $\mathbf{U}$ and $\mathbf{V}$ with standard-deviation matrices $\mathbf{D}_U$ and $\mathbf{D}_V$, the cross-correlation matrix is $\mathbf{D}_U^{-1} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{D}_V^{-1}$.

Some books write $\mathbf{P}_\rho$ for the correlation matrix. We use $\mathbf{R}$ in this chapter, and $\hat{\mathbf{R}}$ for its sample counterpart in [Section 2.7](#).

Proposition 2.4.1 — Properties of correlation matrices

Let $\mathbf{R}$ be the correlation matrix of $\mathbf{Y}$.

- (a) $\mathbf{R}$ is the covariance matrix of the standardized vector $\mathbf{Z} = \mathbf{D}_\sigma^{-1}(\mathbf{Y} - \boldsymbol{\mu})$. Hence $\mathbf{R}$ is nonnegative definite with unit diagonal, and $\text{tr}(\mathbf{R}) = p$.
- (b) $|\rho_{ij}| \leq 1$, with $|\rho_{ij}| = 1$ iff $Z_j = \rho_{ij} Z_i$ with probability one.
- (c) $\text{rank}(\mathbf{R}) = \text{rank}(\mathbf{\Sigma})$. In particular, $\mathbf{R}$ is positive definite iff $\mathbf{\Sigma}$ is.
- (d) If $\mathbf{Y}^* = \mathbf{D}\mathbf{Y} + \mathbf{c}$ with $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$ and every $d_i \neq 0$, then $\rho_{ij}^* = \text{sign}(d_i d_j) \rho_{ij}$. Positive rescaling and shifting leave $\mathbf{R}$ unchanged.
- (e) Conversely, every nonnegative definite matrix with unit diagonal is the correlation matrix of some random vector.

Proof.

- (a) Theorem 2.2.1 gives $\text{Cov}(\mathbf{Z}) = \mathbf{D}_\sigma^{-1} \mathbf{\Sigma} \mathbf{D}_\sigma^{-1} = \mathbf{R}$, and Theorem 2.2.3(a) applies.
- (b) The 2×2 matrix of rows and columns i, j of $\mathbf{R}$ is the covariance matrix of (Z_i, Z_j) , so it is nonnegative definite and its determinant $1 - \rho_{ij}^2$ is nonnegative. If $\rho_{ij} = \pm 1$, the vector $(\rho_{ij}, -1)^\top$ is in the null space of that 2×2 matrix, so $\rho_{ij} Z_i - Z_j = 0$ with probability one by Theorem 2.2.3(c). Conversely, $Z_j = c Z_i$ with $\text{Var}(Z_i) = \text{Var}(Z_j) = 1$ forces $c = \pm 1$ and $\rho_{ij} = c$.
- (c) $\mathbf{D}_\sigma$ is nonsingular.
- (d) $\text{Cov}(\mathbf{Y}^*) = \mathbf{D} \mathbf{\Sigma} \mathbf{D}$ has entries $d_i d_j \sigma_{ij}$ and standard deviations $|d_i| \sigma_{ii}^{1/2}$.
- (e) Such a matrix is a covariance matrix by Theorem 2.2.3(b), and its variances are one.

■

Part (d) is the reason correlation is the natural scale-free summary. Part (e) is less obvious than it looks. A symmetric matrix with unit diagonal and off-diagonal entries in $[-1, 1]$ need not be

a correlation matrix, because the correlations must be jointly consistent.

Proposition 2.4.2 — Three correlations

Let $|\rho_{12}| < 1$ and $|\rho_{13}| < 1$. The matrix

$$\mathbf{R} = \begin{pmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{pmatrix}$$

is a correlation matrix iff

$$\rho_{12}\rho_{13} - \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)} \leq \rho_{23} \leq \rho_{12}\rho_{13} + \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}.$$

Proof. By Proposition 2.4.1(e) we must decide when $\mathbf{R}$ is nonnegative definite. Write $\mathbf{r} = (\rho_{12}, \rho_{13})^\top$ and let $\mathbf{R}_2$ be the lower right 2×2 block. The nonsingular matrix $\mathbf{T} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ -\mathbf{r} & \mathbf{I}_2 \end{pmatrix}$ gives

$$\mathbf{T}\mathbf{R}\mathbf{T}^\top = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top \end{pmatrix},$$

and $\mathbf{a}^\top \mathbf{R} \mathbf{a} = \mathbf{b}^\top (\mathbf{T}\mathbf{R}\mathbf{T}^\top) \mathbf{b}$ with $\mathbf{b} = (\mathbf{T}^\top)^{-1} \mathbf{a}$ ranges over all vectors as $\mathbf{a}$ does. So $\mathbf{R}$ is nonnegative definite iff the Schur complement

$$\mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top = \begin{pmatrix} 1 - \rho_{12}^2 & \rho_{23} - \rho_{12}\rho_{13} \\ \rho_{23} - \rho_{12}\rho_{13} & 1 - \rho_{13}^2 \end{pmatrix}$$

is. A symmetric 2×2 matrix with positive diagonal is nonnegative definite iff its determinant is nonnegative, because its eigenvalues have positive sum and their product is the determinant. The determinant is $(1 - \rho_{12}^2)(1 - \rho_{13}^2) - (\rho_{23} - \rho_{12}\rho_{13})^2$, and it is nonnegative exactly on the stated interval. ■

If Y_2 and Y_3 both correlate 0.8 with Y_1 , the interval is $[0.28, 1]$. Two variables that are strongly related to a third cannot be negatively correlated with each other, nor even uncorrelated. The quantity $(\rho_{23} - \rho_{12}\rho_{13}) / \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}$, which the proof forces into $[-1, 1]$, is the correlation of Y_2 and Y_3 after the linear effect of Y_1 has been removed from both. Section 2.5 develops this residualization, and Definition 3.6.1 names the result the partial correlation.

Correlation matrices do not transform simply under linear maps: the correlation matrix of $\mathbf{A}\mathbf{Y}$ must be recomputed from $\mathbf{A}\mathbf{\Sigma}\mathbf{A}^\top$. Theory is therefore done with covariance matrices, and correlation matrices are used for description.

2.4.1 Exercises

2.4.1.1 A. Check your understanding

Exercise A1.

Show that $(1 - \rho)\mathbf{I}_p + \rho\mathbf{1}\mathbf{1}^\top$ is a correlation matrix iff $-1/(p - 1) \leq \rho \leq 1$. Explain the

lower bound through the variance of $\sum_i Z_i$.

Solution. The matrix sends $\mathbf{1}$ to $(1 + (p-1)\rho)\mathbf{1}$ and every $\mathbf{u} \perp \mathbf{1}$ to $(1 - \rho)\mathbf{u}$. Its eigenvalues are $1 + (p-1)\rho$ (once) and $1 - \rho$ ($p-1$ times). By Proposition 2.4.1(e) it is a correlation matrix iff both are nonnegative. If $\mathbf{Z}$ has this correlation matrix, $\text{Var}(\sum_i Z_i) = \mathbf{1}^\top \mathbf{R} \mathbf{1} = p[1 + (p-1)\rho]$, which must be nonnegative. Many variables cannot all be strongly negatively correlated with one another, because their sum would then have negative variance.

2.4.1.2 B. Practice

Exercise B1.

Show that $\det(\mathbf{\Sigma}) \leq \prod_i \sigma_{ii}$ for every covariance matrix, with equality for positive definite $\mathbf{\Sigma}$ iff $\mathbf{\Sigma}$ is diagonal. *Hint:* factor out $\mathbf{D}_\sigma$ and use the eigenvalues of $\mathbf{R}$.

Solution. If some $\sigma_{ii} = 0$, the i th row of $\mathbf{\Sigma}$ vanishes (apply Theorem 2.2.3(c) with $\mathbf{a}$ the i th coordinate vector) and both sides are zero. Otherwise $\det \mathbf{\Sigma} = \det(\mathbf{D}_\sigma)^2 \det \mathbf{R} = \prod_i \sigma_{ii} \cdot \det \mathbf{R}$. The eigenvalues $\lambda_i \geq 0$ of $\mathbf{R}$ sum to $\text{tr} \mathbf{R} = p$, so by the inequality of arithmetic and geometric means $\det \mathbf{R} = \prod_i \lambda_i \leq (p^{-1} \sum_i \lambda_i)^p = 1$. Equality holds iff every $\lambda_i = 1$, that is, iff $\mathbf{R} = \mathbf{I}$, that is, iff $\mathbf{\Sigma}$ is diagonal.

2.4.1.3 C. Going deeper

Exercise C1.

- (a) Show that a $p \times p$ matrix is a correlation matrix iff it is the Gram matrix $(\mathbf{u}_i^\top \mathbf{u}_j)$ of p unit vectors in $\mathbb{R}^p$. (b) For $p = 3$, write $\rho_{ij} = \cos \vartheta_{ij}$ with $\vartheta_{ij} \in [0, \pi]$. Show that Proposition 2.4.2 is equivalent to the spherical triangle inequalities $|\vartheta_{12} - \vartheta_{13}| \leq \vartheta_{23} \leq \min(\vartheta_{12} + \vartheta_{13}, 2\pi - \vartheta_{12} - \vartheta_{13})$.